

FACTSHEET FS08 · Short-term reversal

Buy last month's losers, sell last month's winners

Henrik Gjerner · Rude Investment Consulting · Project 10 sidebar: strategy factsheets · data to 31 August 2026

STRATEGY One-month reversal (liquidity provision)	ORIGIN Jegadeesh (1990); Lehmann (1990)	FAMILY Reversal & bottom-fishing
UNIVERSES US · EU · UK · DK · SCANDI · World (point-in-time)	BACKTEST Feb 2013 – Aug 2026, monthly, net of costs	BOOKS Long/short and long-only (last month's losers)
REBALANCE Monthly	STATUS Descriptive factsheet, not a registered trial	

The claim. Stocks with the worst returns last month outperform those with the best returns next month. Jegadeesh (1990) and Lehmann (1990) documented it for US stocks; it is large before costs and famously expensive to trade. **What we find, 2013–2026, point-in-time universes, after costs:** the long/short earned -9.7% to +0.1% a year and was positive in 1 of the six; no universe passes the 2.87 gate. The long-only book (last month's losers) had a higher Sharpe than the equal-weight universe in none of the six, with alphas of -7.1% to -0.8%, none past the gate. **What goes wrong (§10):** the main drags are costs of about 4.0% a year, a weaker second half (2020–26). None of the pre-declared fixes passes the gate in both windows. **Classification (§11):** Reversal & bottom-fishing; its returns sit closest to the momentum cluster of the signal library. **360° view (§12):** nearest neighbours 55-day breakout, Net tail (MAX+MIN); after its closest neighbours and the market it keeps an alpha of -4.5% to +5.7% (largest |t| 2.2), so it carries some information of its own.

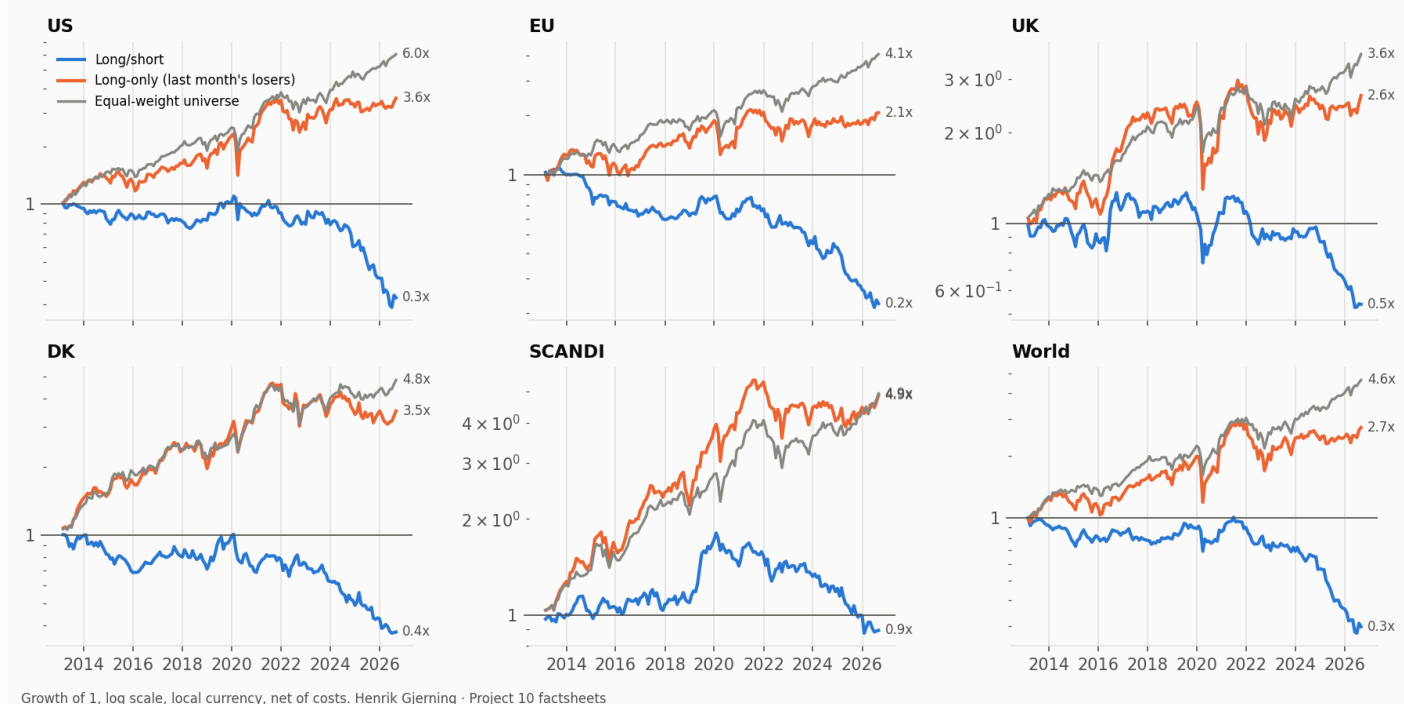
1. Headline performance

Universe	Names / groups	L/S return / yr	L/S t	L/S CAPM alpha (t)	L/S beta	Long-only CAGR	Long-only Sharpe	Long-only max DD	EW CAGR	EW Sharpe	Size-weighted CAGR
US	503 / 10	-6.6%	-1.58	-12.1% (-2.7)	0.39	+9.8%	0.53	-39%	+14.2%	0.96	+15.3%
EU	238 / 10	-9.7%	-2.25	-13.1% (-3.0)	0.30	+5.5%	0.37	-33%	+10.9%	0.79	+10.1%
UK	242 / 10	-3.0%	-0.58	-7.2% (-1.4)	0.40	+7.4%	0.44	-49%	+9.9%	0.74	+8.5%
DK	19 / 3	-5.9%	-1.61	-7.4% (-1.9)	0.12	+9.7%	0.58	-35%	+12.3%	0.82	+10.3%
SCANDI	62 / 5	+0.1%	0.02	-1.0% (-0.2)	0.08	+12.4%	0.76	-34%	+12.5%	0.90	+10.7%

Universe	Names / groups	L/S return / yr	L/S t	L/S CAPM alpha (t)	L/S beta	Long-only CAGR	Long-only Sharpe	Long-only max DD	EW CAGR	EW Sharpe	Size-weighted CAGR
World	975 / 10	-7.6%	-1.85	-12.6% (-3.1)	0.39	+7.7%	0.45	-42%	+12.0%	0.80	+10.8%

Monthly, local currency (World in USD). L/S = equal-weighted last month's losers minus last month's winners, net of 10 bp per side on turnover, a size-tiered borrow fee on the short leg and, where the book is net long cash, financing at the USD risk-free rate + 0.5%. Newey-West t (6 lags). CAPM vs the equal-weight universe. Size-weighted universe = market-cap weights in the US, traded-value weights elsewhere (a proxy for the index).

Short-term reversal on six stock universes, 2013-2026



2. Strategy description

Stocks with the worst returns last month outperform those with the best returns next month. Jegadeesh (1990) and Lehmann (1990) documented it for US stocks; it is large before costs and famously expensive to trade.

Why it might work. Compensation for providing liquidity to investors who trade on non-information (Nagel 2012); overreaction to short-term news.

Rule	Original (Jegadeesh (1990); Lehmann (1990))	This factsheet
Signal	Previous month's (or week's) return	Return in month t
Portfolio	Decile L/S, one month	Extreme quantile L/S, one month
Weighting	Equal or value weighted	Equal weighted

3. Data load

Universe	Prices	Membership	Names eligible (median)	Currency
US	Sharadar SEP <code>closeadj</code> (total return), 805 tickers incl. delisted	Point-in-time: top 500 by market cap each month-end (Sharadar filings × price, A06 method)	~499	USD
EU	Project1 yfinance cache, Adj Close (TR)	Point-in-time: union of 11 national blue-chip indices (CAC 40, DAX, AEX, IBEX 35, FTSE MIB, OMXS30, OMXC25, OMXH25, BEL 20, PSI-20, WIG20), quarterly Wikipedia snapshots 2012–2026; 83% of member-quarters priced	~238	local (EUR, SEK, DKK, PLN)
UK	Project1 cache, Close + pence dividends (Yahoo under-adjusts LSE ~100×)	Point-in-time: FTSE 100 + FTSE 250, quarterly snapshots; 75% priced	~242	GBP
DK	Project1 cache, Adj Close	Point-in-time: OMX Copenhagen 25, quarterly snapshots; 91% priced	~19	DKK
SCANDI	Project1 cache, Adj Close	Point-in-time: OMXC25 + OMXS30 + OMXH25 (Oslo has no snapshot history); 88% priced	~62	local (DKK, SEK, EUR)
World	US Sharadar + UK + EU above	Point-in-time union of the three; local-currency returns summed, not converted	~963	mixed

Membership comes from Project2's cache of dated Wikipedia index pages (the same parsing as `run_wiki_pit.py` / `run_eu_pit.py`), with a short list of verified ticker renames (e.g. *NZYM-B* → *NSIS-B*, *WDH* → *DEMANT*, *DAI* → *MBG*). A name is eligible from the day after the snapshot that lists it, once it has 60 days of prices. Member-quarters without a price are mostly delisted names that Yahoo no longer serves, so some survivorship remains; see §13.

4. Signal creation

$R_{1,i,t}$ = total return of stock *i* in month *t*. Signals use only information up to month-end *t* and are traded in month *t*+1.

Eligibility: in the point-in-time universe on any of the last five trading days of month *t*.

5. Model build

- Sort eligible stocks into equal-count groups (deciles ≥ 100 names, quintiles 50–99, terciles < 50; fixed per universe).
- **Long/short** = last month's losers minus last month's winners; **long-only** = last month's losers; benchmarks: equal-weight universe and size-weighted universe.
- Equal weights within each leg, rebalanced monthly.

- Costs: 10 bp per side on the one-way turnover of each leg; short-leg borrow fee 0.25% / 0.75% / 2% a year by size tier (largest 50% / next 30% / smallest 20% of the universe, `code/borrow.py`); net long cash financed at the USD risk-free rate + 0.5%. Code: `code/xs.py` (engine), `code/analyse_xs.py`, `code/build_xs.py`.

6. Performance in detail

US (top 500, point-in-time), median 503 names, 10 groups

Book	Mean / yr	CAGR	Vol	Sharpe [95% CI]	Sortino	Max DD	Hit	t(mean)	Beta	Alpha/yr (t)	CVaR 95% (month)
L/S (net)	-6.6%	-7.9%	18.2%	-0.36 [-0.85, 0.12]	-0.57	-74%	40%	-1.58	0.39	-12.1% (-2.7)	12.7%
Long-only (net)	+12.0%	+9.8%	22.5%	0.53 [0.14, 1.03]	0.68	-39%	61%	2.64	1.32	-7.1% (-3.1)	13.2%
Last month's losers (gross)	+14.0%	+12.1%	22.5%	0.62 [0.22, 1.14]	0.85	-39%	64%	3.10	1.32	-5.1% (-2.2)	13.0%
Last month's winners (gross)	+15.7%	+15.2%	17.0%	0.92 [0.37, 1.52]	1.52	-28%	63%	3.47	0.93	+2.2% (0.8)	9.8%
EW universe	+14.5%	+14.2%	15.1%	0.96 [0.52, 1.53]	1.52	-25%	67%	4.79	1.00	–	8.9%
Size-weighted universe	+15.4%	+15.3%	14.4%	1.07 [0.61, 1.64]	1.79	-24%	69%	5.27	0.90	+2.3% (1.8)	8.3%

EU (11 national blue-chip indices, point-in-time), median 238 names, 10 groups

Book	Mean / yr	CAGR	Vol	Sharpe [95% CI]	Sortino	Max DD	Hit	t(mean)	Beta	Alpha/yr (t)	CVaR 95% (month)
L/S (net)	-9.7%	-10.4%	15.9%	-0.61 [-1.11, -0.08]	-0.81	-80%	44%	-2.25	0.30	-13.1% (-3.0)	10.7%
Long-only (net)	+7.3%	+5.5%	19.9%	0.37 [-0.07, 0.85]	0.42	-33%	58%	1.54	1.20	-6.4% (-2.6)	11.8%
Last month's losers (gross)	+9.3%	+7.6%	19.9%	0.47 [0.03, 0.96]	0.60	-32%	58%	1.97	1.20	-4.4% (-1.8)	11.6%
Last month's winners (gross)	+14.1%	+13.7%	15.6%	0.91 [0.41, 1.46]	1.53	-25%	60%	3.61	0.91	+3.8% (1.5)	8.6%

Book	Mean / yr	CAGR	Vol	Sharpe [95% CI]	Sortino	Max DD	Hit	t(mean)	Beta	Alpha/yr (t)	CVaR 95% (month)
EW universe	+11.4%	+10.9%	14.5%	0.79 [0.33, 1.33]	1.22	-26%	64%	3.35	1.00	–	8.3%
Size-weighted universe	+10.7%	+10.1%	14.3%	0.75 [0.28, 1.28]	1.14	-26%	60%	3.12	0.95	-0.2% (-0.2)	8.3%

UK (FTSE 350, point-in-time), median 242 names, 10 groups

Book	Mean / yr	CAGR	Vol	Sharpe [95% CI]	Sortino	Max DD	Hit	t(mean)	Beta	Alpha/yr (t)	CVaR 95% (month)
L/S (net)	-3.0%	-4.5%	17.6%	-0.17 [-0.82, 0.40]	-0.36	-58%	48%	-0.58	0.40	-7.2% (-1.4)	10.1%
Long-only (net)	+9.7%	+7.4%	22.1%	0.44 [-0.06, 1.10]	0.50	-49%	53%	1.63	1.36	-4.6% (-1.3)	12.8%
Last month's losers (gross)	+11.8%	+9.6%	22.1%	0.53 [0.03, 1.21]	0.66	-48%	55%	1.96	1.36	-2.6% (-0.7)	12.6%
Last month's winners (gross)	+10.0%	+9.0%	16.4%	0.61 [0.16, 1.13]	0.86	-26%	60%	2.74	0.96	-0.1% (-0.1)	9.7%
EW universe	+10.5%	+9.9%	14.2%	0.74 [0.26, 1.35]	1.08	-30%	59%	3.24	1.00	–	8.8%
Size-weighted universe	+9.0%	+8.5%	12.7%	0.71 [0.21, 1.31]	1.01	-27%	62%	2.94	0.82	+0.4% (0.2)	7.7%

DK (OMXC25, point-in-time), median 19 names, 3 groups

Book	Mean / yr	CAGR	Vol	Sharpe [95% CI]	Sortino	Max DD	Hit	t(mean)	Beta	Alpha/yr (t)	CVaR 95% (month)
L/S (net)	-5.9%	-7.0%	15.8%	-0.38 [-0.85, 0.10]	-0.56	-63%	47%	-1.61	0.12	-7.4% (-1.9)	10.5%
Long-only (net)	+11.3%	+9.7%	19.6%	0.58 [0.09, 1.21]	0.72	-35%	61%	2.38	1.13	-3.3% (-1.7)	13.2%
Last month's losers (gross)	+12.8%	+11.4%	19.6%	0.66 [0.16, 1.30]	0.86	-34%	61%	2.70	1.13	-1.8% (-0.9)	13.1%
Last month's winners (gross)	+14.9%	+14.1%	18.3%	0.81 [0.31, 1.37]	1.29	-32%	63%	3.24	1.02	+1.8% (0.8)	9.7%
EW universe	+12.9%	+12.3%	15.6%	0.82 [0.30, 1.44]	1.26	-33%	64%	3.21	1.00	–	9.3%

Book	Mean / yr	CAGR	Vol	Sharpe [95% CI]	Sortino	Max DD	Hit	t(mean)	Beta	Alpha/yr (t)	CVaR 95% (month)
Size-weighted universe	+11.4%	+10.8%	15.0%	0.76 [0.29, 1.33]	1.14	-28%	63%	3.23	0.94	-0.4% (-0.5)	9.0%

Sharpe CI: stationary bootstrap (mean block 6 months, 5,000 draws). Max drawdown on monthly returns.

Calendar-year returns, L/S (net)

Year	US	EU	UK	DK	SCANDI	World
2013	-4.8%	+0.1%	-2.5%	-0.2%	+0.4%	-8.1%
2014	-14.5%	-18.2%	-8.3%	-17.8%	+0.1%	-17.1%
2015	+7.6%	-10.7%	-2.3%	-16.6%	+5.0%	+6.0%
2016	-1.5%	-13.0%	+30.6%	+19.5%	+7.2%	+5.2%
2017	-6.0%	-6.4%	-3.8%	-2.0%	-4.7%	-8.2%
2018	+1.8%	+4.4%	+3.7%	-11.5%	+1.5%	+1.6%
2019	+24.5%	+24.8%	-5.7%	+40.5%	+57.7%	+12.3%
2020	-14.6%	-16.0%	+0.7%	-31.2%	-17.5%	-3.5%
2021	+3.0%	+3.4%	+1.2%	+13.8%	+7.6%	+4.9%
2022	-13.0%	-17.7%	-19.3%	-4.2%	-8.1%	-22.7%
2023	-0.6%	-19.0%	+6.1%	-16.3%	-6.2%	-0.8%
2024	-22.4%	-4.5%	-7.2%	-21.6%	-8.4%	-17.6%
2025	-32.0%	-38.5%	-25.6%	-12.7%	-17.8%	-36.1%
2026	-21.0%	-15.0%	-16.5%	-12.5%	-10.1%	-17.9%

Calendar-year returns, long-only (net)

Year	US	EU	UK	DK	SCANDI	World
2013	+30.2%	+25.2%	+21.9%	+45.8%	+27.3%	+24.4%
2014	+2.8%	-11.4%	-4.3%	+4.5%	+12.9%	-8.3%
2015	-3.8%	+0.5%	-3.7%	+11.8%	+12.0%	-1.1%
2016	+14.2%	+6.7%	+63.9%	+18.3%	+32.0%	+15.0%
2017	+11.1%	+17.6%	+26.1%	+17.5%	+14.2%	+20.2%
2018	-5.8%	-2.2%	-7.8%	-16.9%	-9.1%	-7.1%
2019	+47.9%	+37.6%	+17.7%	+51.7%	+73.8%	+37.1%
2020	+12.7%	-8.2%	-10.0%	+17.3%	+11.9%	+11.7%
2021	+35.5%	+20.0%	+25.5%	+32.1%	+25.3%	+27.8%
2022	-29.5%	-17.1%	-24.9%	-19.6%	-19.6%	-32.7%
2023	+27.0%	+5.4%	+15.4%	+1.7%	+3.7%	+27.1%

Year	US	EU	UK	DK	SCANDI	World
2024	+3.2%	+2.0%	-3.1%	-9.9%	-8.5%	-1.7%
2025	+3.4%	-1.5%	+2.3%	+3.6%	+9.2%	+3.2%
2026	+7.4%	+13.3%	+8.3%	+0.1%	+9.1%	+11.0%

7. Trading record

The rebalance log per universe is in `results/xs_FS08_record_<universe>.csv` (names per leg, turnover, leg returns, gross and net L/S).

Universe	Rebalances	Names long / short	Turnover long leg / month	Turnover short leg / month	L/S months positive	Worst L/S month	When
US	163	51 / 50	86%	88%	40%	-22.8%	2020-03
EU	163	24 / 23	85%	87%	44%	-13.3%	2024-02
UK	163	25 / 24	84%	87%	48%	-22.4%	2020-03
DK	163	7 / 6	65%	69%	47%	-14.2%	2020-02
SCANDI	163	13 / 12	79%	80%	52%	-11.9%	2026-01
World	163	98 / 97	85%	87%	42%	-20.3%	2020-03

Current book (signal at 31 August 2026; first 8 names per leg)

Universe	Long: last month's losers	Short: last month's winners
US	DVA, EIX, PCG, APP, APTV, TPR, NRG, WDC	MRNA, PLTR, VEEV, CRM, PSKY, NEM, NOW, SMC
EU	BPOST.BR, ZAL.DE, RBREW.CO, ANA.MC, FME.DE, FER.MC, MRL.MC, GLE.PA	QTCOM.HE, JSW.WA, MAERSK-B.CO, MAERSK-A.CO, AGFB.BR, VWS.CO, BAVA.CO, SAP.DE
UK	GBG.L, TRN.L, RPI.L, BATS.L, PRN.L, IMB.L, CCH.L, PRU.L	ONT.L, HOC.L, PAF.L, KNOS.L, HAS.L, HWG.L, EDV.L, FRES.L
DK	RBREW.CO, CARL-B.CO, ORSTED.CO, DSV.CO, NOVO-B.CO, AMBU-B.CO, ZEAL.CO, GN.CO	MAERSK-B.CO, MAERSK-A.CO, VWS.CO, BAVA.CO, GMAB.CO, NSIS-B.CO, COLO-B.CO, JYSK.CO
SCANDI	RBREW.CO, CARL-B.CO, ORSTED.CO, VOLV-B.ST, WRT1V.HE, AZN.ST, DSV.CO, NOVO-B.CO	QTCOM.HE, MAERSK-B.CO, MAERSK-A.CO, VWS.CO, BAVA.CO, BOL.ST, EVO.ST, GMAB.CO

8. Risk and factor attribution

JKP 7 themes (US, UK, DK, World) or French Europe 5F + WML (EU, SCANDI); Newey-West t, 6 lags.

Long/short

| Universe | Factor model | Alpha / yr | t(alpha) | R² | Loadings with |t| ≥ 2 (t) | ---|---|---|---|---|---| | US | JKP USA 7 themes | -5.3% | -1.44 | 0.56 | short_term_reversal +2.84 (+11.1) | | EU | French Europe 5F + WML | -5.7% | -1.44 | 0.15 | Mkt-RF +0.20 (+2.0), SMB +0.52 (+2.8), WML -0.33 (-2.4) | | UK | JKP GBR 7 themes | -2.8% | -0.72 | 0.50 | mkt +0.18 (+3.0), short_term_reversal +1.99 (+6.2) | | DK | JKP DNK 7 themes | -0.6% | -0.16 | 0.33 | momentum -0.35 (-2.5), short_term_reversal +0.88 (+7.4) | | SCANDI | French Europe 5F + WML | +2.2% | 0.61 | 0.06 | none |t| ≥ 2 | | World | JKP World 7 themes | -11.8% | -2.59 | 0.45 | mkt +0.23 (+3.1), quality -0.41 (-2.2), short_term_reversal +1.85 (+6.2) |

Long-only

| Universe | Factor model | Alpha / yr | t(alpha) | R² | Loadings with |t| ≥ 2 (t) | ---|---|---|---|---|---| | US | JKP USA 7 themes | +1.4% | 0.69 | 0.90 | mkt +0.97 (+17.3), size -0.23 (-2.6), low_risk -0.65 (-3.4), short_term_reversal +1.53 (+7.4) | | EU | French Europe 5F + WML | +5.3% | 1.73 | 0.67 | Mkt-RF +0.86 (+10.1), WML -0.39 (-4.5) | | UK | JKP GBR 7 themes | +3.7% | 0.98 | 0.81 | mkt +0.67 (+14.0), low_risk -1.30 (-5.5), short_term_reversal +1.14 (+4.2) | | DK | JKP DNK 7 themes | +6.8% | 2.15 | 0.73 | mkt +0.76 (+10.6), value +0.16 (+2.0), momentum -0.28 (-2.7), low_risk -0.47 (-3.8), short_term_reversal +0.48 (+4.4) | | SCANDI | French Europe 5F + WML | +10.1% | 3.04 | 0.56 | Mkt-RF +0.75 (+9.2), RMW +0.57 (+2.3), WML -0.30 (-3.0) | | World | JKP World 7 themes | +0.6% | 0.19 | 0.85 | mkt +0.85 (+13.3), low_risk -0.39 (-3.0), short_term_reversal +1.07 (+4.7) |

9. Statistical verdict

- **Gate:** |t| > 2.87 (2 books × 6 universes, Bonferroni 0.05/12, two-sided).
- **L/S mean:** positive in 1 of the six; passes in 0 of 6.
- **L/S CAPM alpha:** -13.1% to -1.0%, |t| past the gate in 2 of 6.
- **Long-only:** Sharpe above the equal-weight universe in none of the six; alpha -7.1% to -0.8%, passing in 0 of 6.
- **Probabilistic Sharpe** of the L/S, P(true Sharpe > 0): US 0.10, EU 0.01, UK 0.27, DK 0.08, SCANDI 0.51, World 0.03.
- **Publication decay:** gross L/S +2.2% to +13.4% a year in 2013–19 against -12.0% to -3.9% in 2020–26.

10. What goes wrong, and how to fix it

Method. Diagnose on the design window (2013–2019); apply the same pre-declared fix ladder used for every sort strategy in FS03–FS12; judge on the holdout (Jan 2020 – Aug 2026) with a pooled six-universe paired bootstrap of the Sharpe difference against the previous step. 25 fix trials across FS03–FS12, gate $p < 0.0020$.

10.1 Diagnosis (design window 2013–2019)

Universe	L/S gross / yr	Beta	Beta drag / yr	CAPM alpha (gross)	Costs / yr	Long leg vs EW / yr	EW vs short leg / yr	Turnover / month	Worst months
US	+6.0%	0.32	+4.5%	+1.5%	-4.2%	+1.2%	+4.8%	175%	2014-11 -8%, 2017-05 -7%, 2019-08 -5%
EU	+2.2%	0.22	+2.6%	-0.4%	-4.2%	+1.1%	+1.1%	173%	2015-01 -13%, 2015-09 -9%, 2013-03 -7%
UK	+7.3%	0.09	+1.2%	+6.1%	-4.1%	+3.2%	+4.1%	171%	2016-04 -10%, 2015-07 -9%, 2016-10 -9%
DK	+5.0%	-0.10	-1.6%	+6.6%	-3.3%	+3.2%	+1.8%	137%	2019-08 -11%, 2014-02 -9%, 2013-05 -8%
SCANDI	+13.4%	0.01	+0.2%	+13.2%	-3.9%	+7.1%	+6.2%	161%	2017-10 -6%, 2017-04 -6%, 2018-06 -6%
World	+3.8%	0.24	+2.9%	+0.9%	-4.2%	+0.9%	+3.0%	174%	2013-03 -8%, 2017-04 -7%, 2014-11 -5%

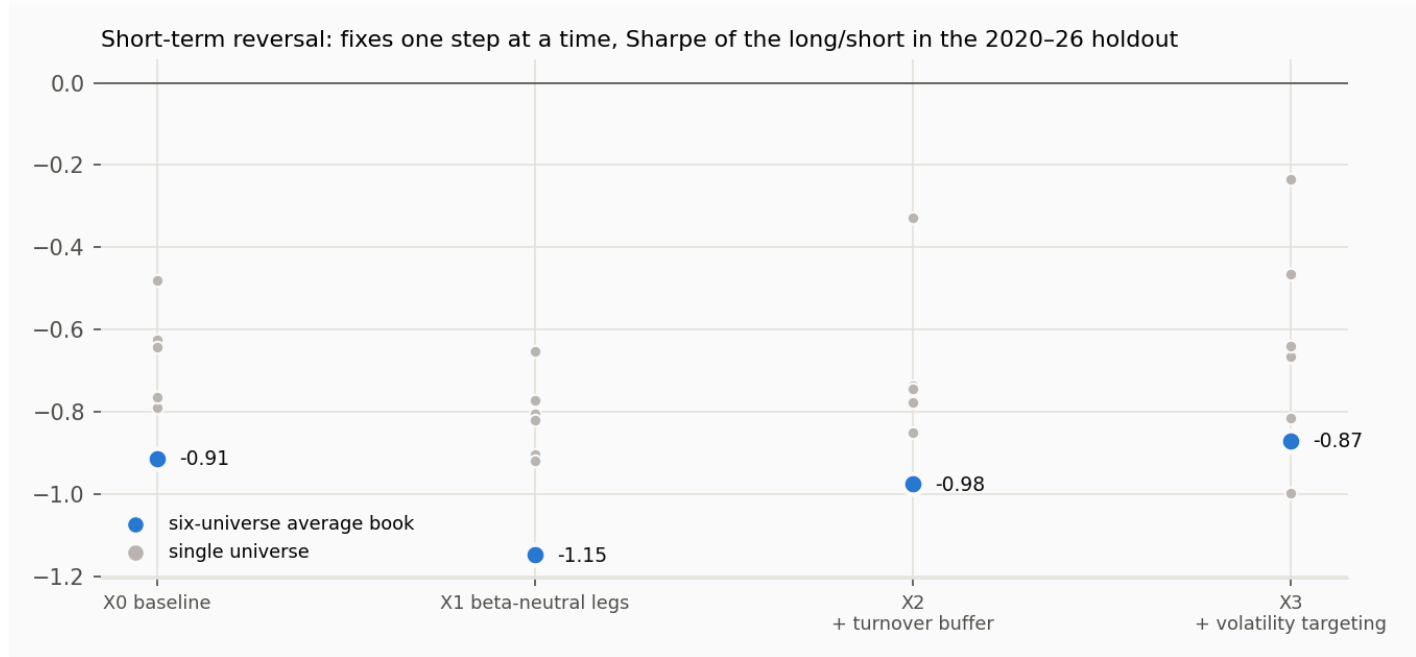
On average across the six universes the gross spread was +6.3% a year, of which the market exposure (beta +0.13) contributed +1.6%; the beta-adjusted spread (CAPM alpha) was +4.6%. Costs took 4.0% a year at 165% monthly turnover across both legs. The long leg beat the universe by +2.8% and the short leg lagged it by +3.5% a year, so most of the spread comes from the short side. Holding longer does not change the picture much: the gross spread 2, 3 and 6 months after formation averages +0.3%, +2.7%, +1.8% a year.

10.2 The fix ladder

X0 baseline → X1 **beta-neutral legs** (each leg scaled by 1/its ex-ante beta) → X2 + **turnover buffer** (enter the extreme 1/q, keep a name until it leaves the extreme 3/q) → X3 + **volatility targeting** (scale the L/S to 10% a year using its trailing 6-month volatility, lagged; Barroso & Santa-Clara 2015; Moreira & Muir 2017).

Step	Design Sharpe (avg)	Δ design (p)	Holdout Sharpe (book)	Δ holdout (p)	Universes improved	Holdout return / yr	Holdout max DD	Holdout alpha vs EW (t)	Costs / yr
X0 baseline	0.11	—	-0.91	—	—	-12.7%	-63%	-16.9% (-3.4)	3.9%
X1 beta-neutral legs	0.01	-0.11 (p 0.838)	-1.15	-0.23 (p 0.880)	1 of 6	-12.5%	-62%	-15.0% (-3.7)	3.8%
X2 + turnover buffer	-0.03	-0.08 (p 0.881)	-0.98	+0.17 (p 0.074)	6 of 6	-10.2%	-57%	-12.7% (-3.3)	3.0%
X3 + volatility targeting	-0.08	-0.12 (p 0.600)	-0.87	+0.11 (p 0.231)	4 of 6	-9.0%	-55%	-11.9% (-3.5)	3.0%

✓ = passes the gate in the holdout. Design Sharpe = average across universes; tests and holdout columns use the six-universe equal-weighted book.



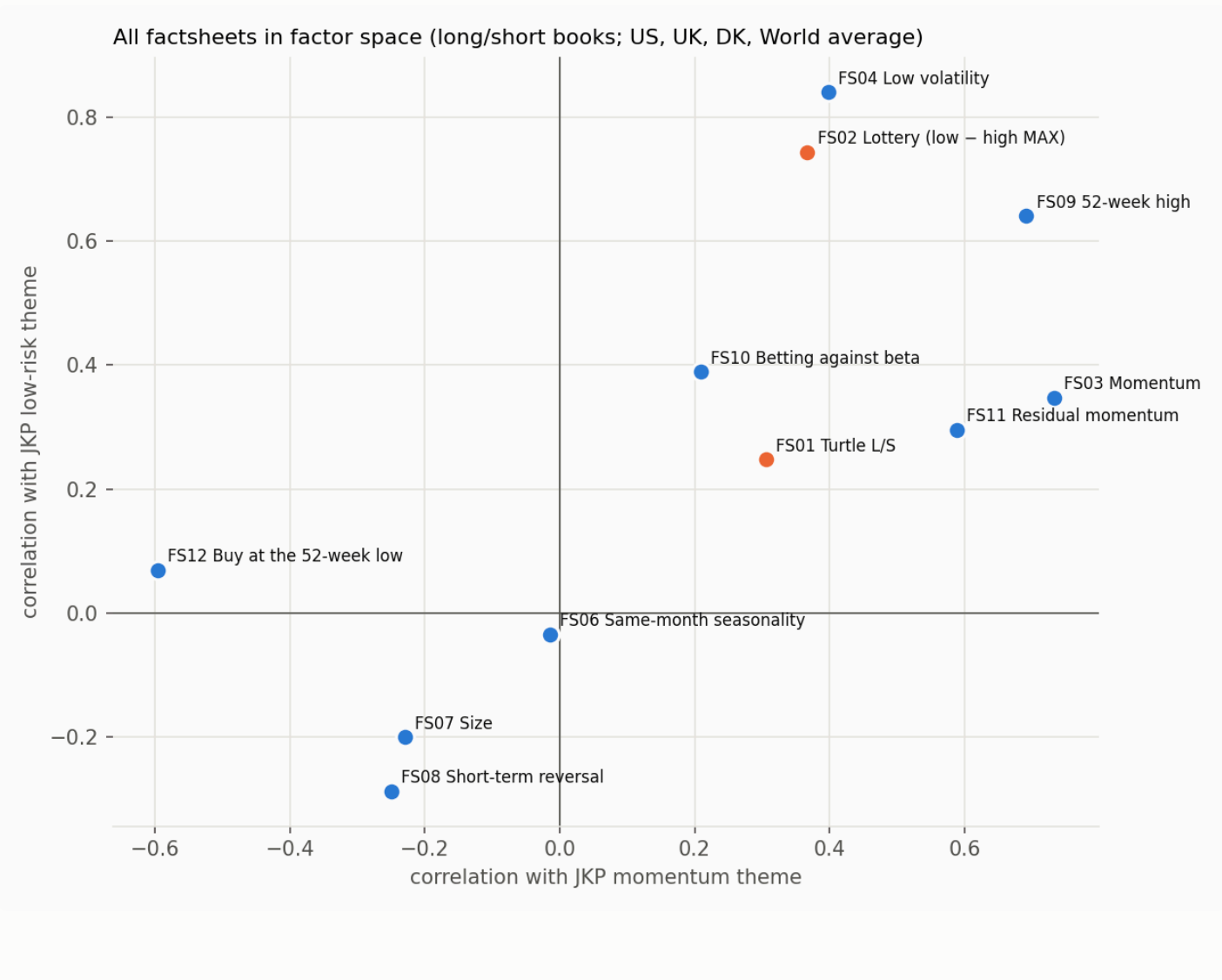
- **X1 beta-neutral legs:** -0.11 design, -0.23 holdout; hurts in both windows.
- **X2 + turnover buffer:** -0.08 design, +0.17 holdout; helps in one window and hurts in the other: not reliable.
- **X3 + volatility targeting:** -0.12 design, +0.11 holdout; helps in one window and hurts in the other: not reliable.

Where it ends: holdout Sharpe -0.91 → -0.87, return -12.7% → -9.0% a year (six-universe book).

11. Classification: where does it fit?

Dimension	Short-term reversal
Series family	Reversal & bottom-fishing
Academic style	One-month reversal (liquidity provision)
Signal	Price only (daily total returns)
Cross-section vs time series	Cross-sectional: stocks ranked against each other each month
Horizon and turnover	One month; 165% of the two legs replaced monthly
Market exposure	L/S beta 0.08 to 0.40
Payoff shape	Treynor–Mazuy γ -2.88 to 0.37 (t -2.5 to 0.5); worst 10% of market months -3.3% to -0.6% a month, best 10% -1.6% to +3.8%
Nearest factor theme	short_term_reversal (3 of 6 universes), WML (2 of 6 universes)
Nearest library signals (returns)	55-day breakout (6 universes), Net tail (MAX+MIN) (6 universes), Near 52-week high (2 universes)
Economic rationale	Compensation for providing liquidity to investors who trade on non-information (Nagel 2012); overreaction to short-term news.

Dimension	Short-term reversal
Publication	Jegadeesh 1990; Lehmann 1990; Nagel 2012 (reversal as liquidity provision)
Where it fits	In the momentum cluster of the library; with some information beyond its neighbours.

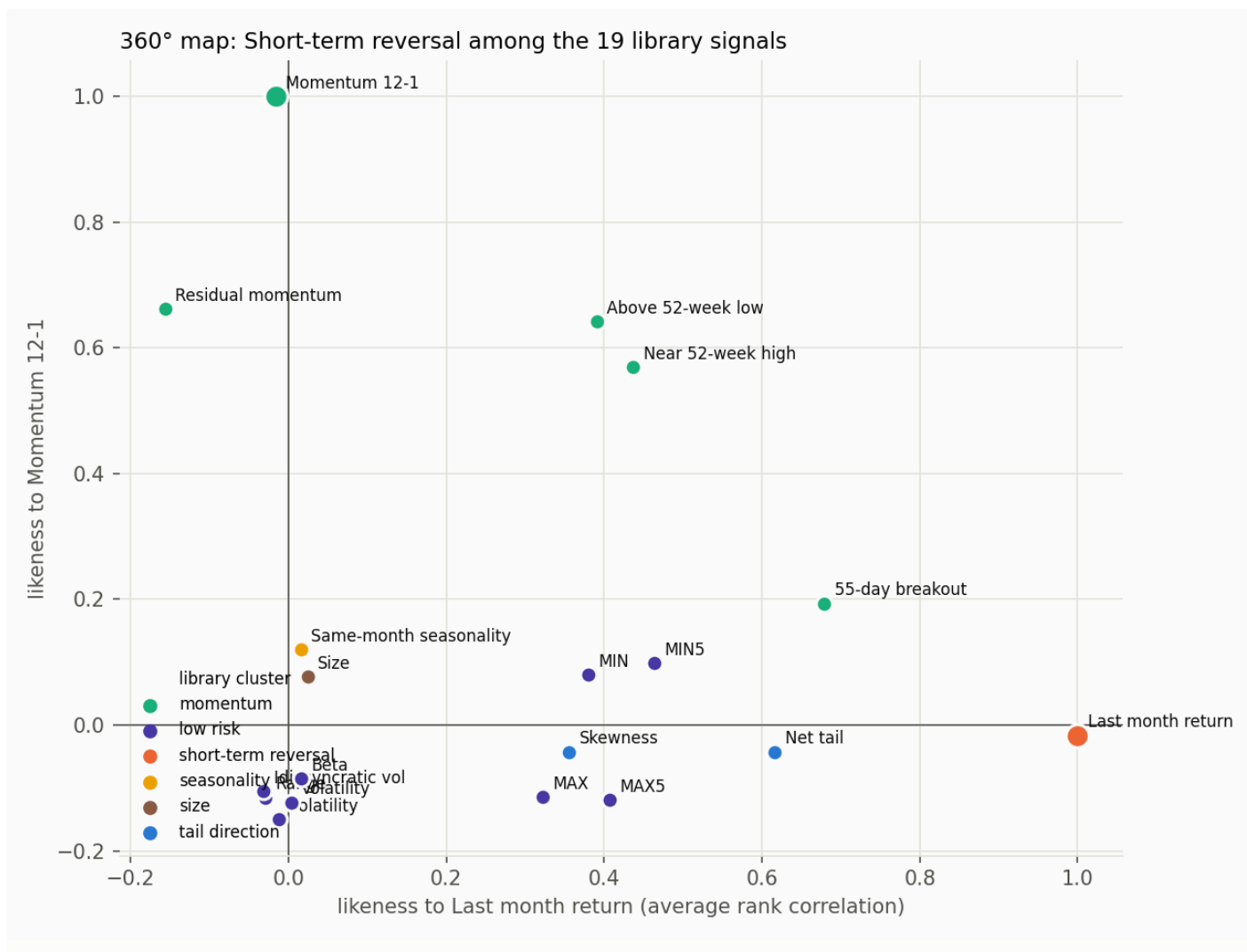


12. 360° view

The signal among the 19 price signals of the shared library (`code/library.py`), on the same universes and months (equal-weighted top minus bottom quantile, gross). Its natural mirror here is **Momentum 12-1**.

Signal (top minus bottom, raw sign)	Cluster	Likeness: stocks	Likeness: returns	Raw L/S, avg of 6	CAPM alpha range	Universes alpha t ≥ +2 / ≤ -2
Last month return (this strategy)	short-term reversal	+1.00	+1.00	+0.9%	-3.2% to +8.4%	0 / 0
55-day breakout	momentum	+0.68	+0.69	+0.7%	+5.6% to +12.6%	3 / 0
Net tail (MAX+MIN)	tail direction	+0.62	+0.62	-1.0%	-5.3% to +4.2%	0 / 0
Near 52-week high	momentum	+0.44	+0.51	+2.5%	+8.3% to +18.8%	6 / 0

Signal (top minus bottom, raw sign)	Cluster	Likeness: stocks	Likeness: returns	Raw L/S, avg of 6	CAPM alpha range	Universes alpha t $\geq +2$ / ≤ -2
MIN5	low risk	+0.46	+0.49	-2.0%	+2.4% to +13.2%	4 / 0
MIN (worst day)	low risk	+0.38	+0.47	-1.5%	+2.9% to +11.0%	3 / 0
Above 52-week low	momentum	+0.39	+0.46	+8.8%	+4.4% to +13.8%	4 / 0
Momentum 12-1 (mirror)	momentum	-0.02	+0.32	+7.7%	+3.9% to +17.8%	5 / 0
Skewness	tail direction	+0.36	+0.27	-0.2%	-5.4% to +4.5%	0 / 1
Range (MAX-MIN)	low risk	-0.03	-0.25	+1.2%	-11.3% to -2.3%	0 / 4
Volatility (252 days)	low risk	+0.00	-0.24	+5.1%	-10.3% to -0.2%	0 / 1
Volatility	low risk	-0.01	-0.24	+3.0%	-10.3% to -3.7%	0 / 5
Beta	low risk	+0.02	-0.22	+5.5%	-15.7% to -2.6%	0 / 3
Size	size	+0.02	+0.21	-1.5%	-3.8% to +4.6%	0 / 0
Idiosyncratic vol	low risk	-0.03	-0.17	-1.5%	-10.4% to -0.8%	0 / 3
Same-month seasonality	seasonality	+0.02	+0.12	-1.8%	-8.5% to +7.2%	0 / 0
MAX5	low risk	+0.41	+0.09	+2.6%	-8.5% to +0.4%	0 / 2
Residual momentum	momentum	-0.16	+0.05	+6.1%	+7.3% to +13.0%	5 / 0
MAX (best day)	low risk	+0.32	+0.01	+1.9%	-9.3% to +1.0%	0 / 2



Spanning. Regressing the L/S on its five closest library neighbours (by return correlation) and the market:

| Universe | Alpha after neighbours / yr | t | R^2 | Neighbours with $|t| \geq 2$ | ---|---|---|---|---| | US | +1.6% | 0.64 | 0.80 | 55-day breakout +0.74 (+7.7), Net tail +0.85 (+8.9) | | EU | +5.7% | 2.05 | 0.67 | 55-day breakout +0.39 (+3.6), Net tail +0.62 (+6.6), MIN5 +0.46 (+3.9), MIN -0.27 (-2.2) | | UK | -3.6% | -1.42 | 0.73 | Net tail +0.74 (+9.3), 55-day breakout +0.59 (+6.2), MIN5 +0.44 (+4.0), Near 52-week high -0.23 (-2.2), MIN -0.44 (-3.4) | | DK | -4.2% | -1.56 | 0.56 | 55-day breakout +0.44 (+5.3), Net tail +0.30 (+4.2), Above 52-week low +0.16 (+2.5) | | SCANDI | -4.5% | -1.58 | 0.51 | 55-day breakout +0.28 (+3.3), Net tail +0.34 (+4.2), Above 52-week low +0.23 (+3.7) | | World | +4.3% | 2.18 | 0.85 | 55-day breakout +0.58 (+7.0), Net tail +1.00 (+11.8), MIN5 +0.39 (+2.9), MIN -0.56 (-3.4) |

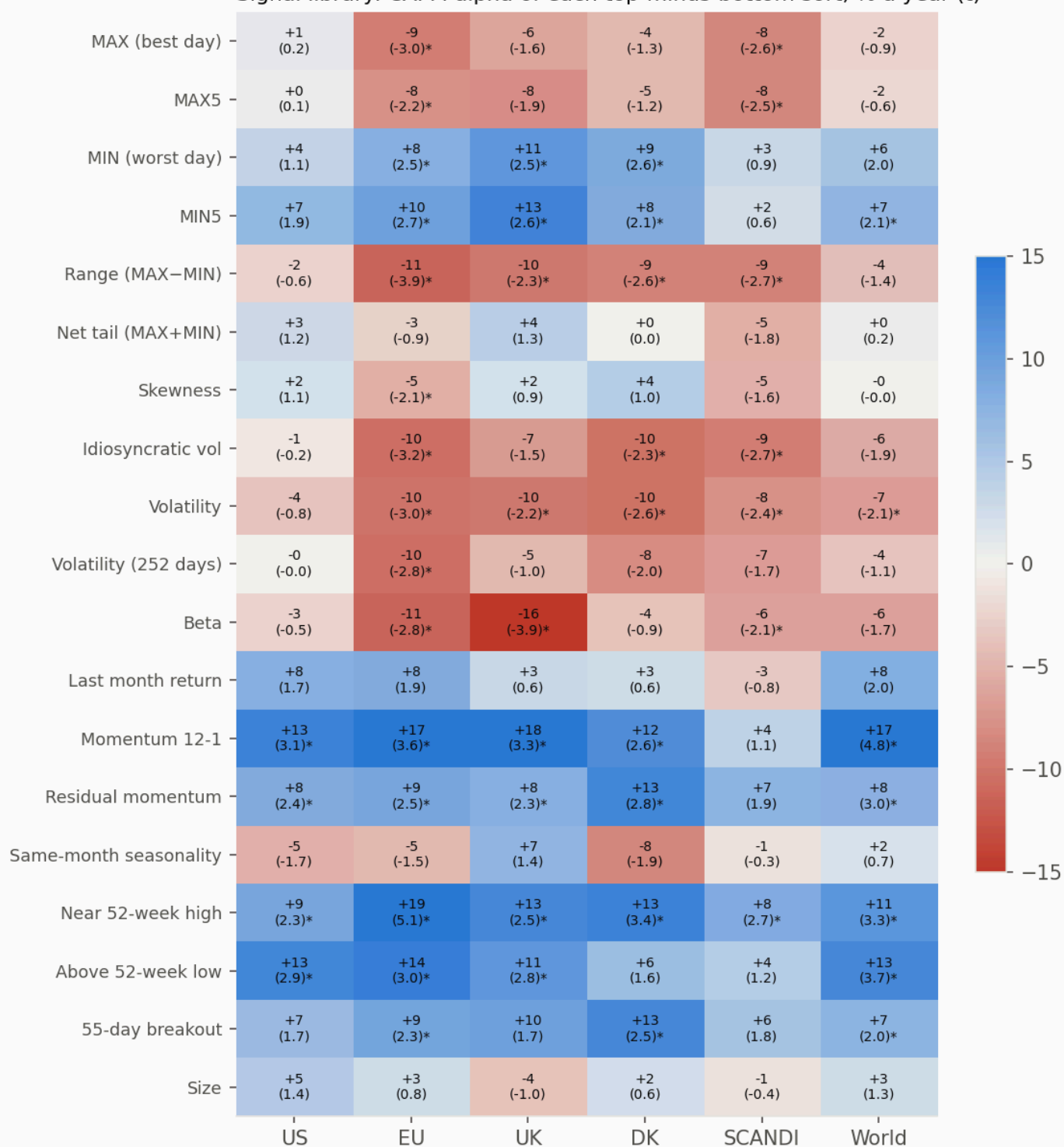
3×3 sort, next-month return a year
(US, EU, UK, SCANDI, World averaged)

low Last month return -	11.6%	11.4%	14.0%
middle -	9.9%	11.8%	14.9%
high Last month return -	10.1%	12.6%	14.2%
	low Momentum 12-1	middle	high Momentum 12-1

Holding Momentum 12-1 fixed (down a column), moving from low to high Last month return changes the return by -1.4, +1.2 and +0.3 points a year; holding Last month return fixed (along a row), moving from low to high Momentum 12-1 changes it by +2.4, +5.0 and +4.1. **Market direction.** Top-minus-bottom return in rising vs falling market months (six-universe average): -4% vs +10% a year. **Horizon:** the gross spread at 1, 2, 3 and 6 months after formation averages +0.9%, +0.3%, +2.7%, +1.8% a year.

The full library, with the CAPM alpha of every signal in every universe:

Signal library: CAPM alpha of each top-minus-bottom sort, % a year (t)



13. Caveats

1. **Residual survivorship.** Point-in-time membership everywhere, but 9–25% of member-quarters outside the US have no price (mostly delisted names); long books are flattered and short books penalised by an unknown amount.
2. **Currency.** Local currency; World sums local-currency returns without conversion.
3. **Equal weighting and flat trading costs.** 10 bp per side for every stock and a size-tiered borrow fee, but no market impact; blue-chip universes only.

4. **Not a registered trial.** Descriptive; the fix ladder is counted inside FS03–FS12 (25 trials).

14. Academic references

- Jegadeesh, N. (1990). Evidence of predictable behavior of security returns. *Journal of Finance*, 45(3), 881–898.
- Lehmann, B. (1990). Fads, martingales, and market efficiency. *Quarterly Journal of Economics*, 105(1), 1–28.
- Nagel, S. (2012). Evaporating liquidity. *Review of Financial Studies*, 25(7), 2005–2039.
- Newey, W. & West, K. (1987). A simple, positive semi-definite, heteroskedasticity and autocorrelation consistent covariance matrix. *Econometrica*, 55(3), 703–708.
- Politis, D. & Romano, J. (1994). The stationary bootstrap. *Journal of the American Statistical Association*, 89(428), 1303–1313.
- Bailey, D. & López de Prado, M. (2012). The Sharpe ratio efficient frontier. *Journal of Risk*, 15(2), 3–44.
- Jensen, T. I., Kelly, B. & Pedersen, L. H. (2023). Is there a replication crisis in finance? *Journal of Finance*, 78(5), 2465–2518.
- Fama, E. & French, K. (2015). A five-factor asset pricing model. *Journal of Financial Economics*, 116(1), 1–22.
- McLean, R. D. & Pontiff, J. (2016). Does academic research destroy stock return predictability? *Journal of Finance*, 71(1), 5–32.
- Novy-Marx, R. & Velikov, M. (2016). A taxonomy of anomalies and their trading costs. *Review of Financial Studies*, 29(1), 104–147.
- Barroso, P. & Santa-Clara, P. (2015). Momentum has its moments. *Journal of Financial Economics*, 116(1), 111–120.
- Moreira, A. & Muir, T. (2017). Volatility-managed portfolios. *Journal of Finance*, 72(4), 1611–1644.
- Treynor, J. & Mazuy, K. (1966). Can mutual funds outguess the market? *Harvard Business Review*, 44(4), 131–136.

15. Reproduce

`code/data.py` → `signals.py`, `signals2.py`, `library.py` → `xs.py` + `analyse_xs.py` FS08 → `build_xs.py`
FS08. Metrics use Project1 `InvestmentLibrary`. Licensed and cached prices are not included.